

Autonomous Emergence Generation System:

Four-Dimension Validation and Semantic Innovation

OEF Team

Multi-Agent Emergence System (MVES) Project

Abstract

This study presents an Autonomous Emergence Generation System (AEGS) that achieves true goal emergence through a four-dimension independence validation framework. Based on the Multi-Objective Self-organizing System (MOSS) framework, the system realizes innovation from predefined behavioral vocabulary to semantic concepts.

Core Contributions:

1. **Four-Dimension Validation Framework:** List, Semantic, Source, and Causal Independence
2. **Vocabulary Expansion:** From 10 to 58 words (5.8x expansion)
3. **Structural Diversification:** 46% 4+ word combinations
4. **Semantic Synthesis:** 45 semantic concepts, 100% semantic goals, 0.936 novelty score

Experimental Results: 478 emergence events, 91.4% success rate, 100% four-dimension validation.

Keywords: Emergent computation, autonomous goal generation, semantic innovation, multi-agent systems

1. Introduction

Emergent computation is an important research direction in artificial intelligence, aiming to produce complex, unpredictable global behaviors through the interaction of simple rules. However, existing systems are mostly limited to predefined goal spaces, making it difficult to achieve true autonomous goal emergence.

Research Motivation:

Traditional multi-agent systems face the following challenges:

- **Limited Goal Space:** Permutations of predefined vocabulary
- **Single Structure:** Fixed-length goal descriptions
- **Lack of Semantic Innovation:** String concatenation rather than concept synthesis

This study aims to address these fundamental issues and achieve true vocabulary innovation and semantic emergence.

2. Methods

2.1 System Architecture

The system is based on the MOSS (Multi-Objective Self-organizing System) framework, including the following core components:

Vocabulary Synthesizer:

- Base vocabulary: 10 predefined behaviors
- Synthesis mechanism: Automatically generates 43 synthetic words
- Total vocabulary space: 58 words (5.8x expansion)

Semantic Synthesizer:

- Predefined rules: 45 semantic synthesis rules
- Dynamic generation: Supports complex semantic synthesis
- Novelty scoring: 0.87-0.97

Goal Discoverer:

- Combination length: 2-5 words
- Selection strategy: Random sampling based on context behaviors
- Semantic validation: Ensures concept validity

2.2 Four-Dimension Validation Framework

List Independence: Verify goal is not in predefined list

Semantic Independence: Verify semantic difference from predefined goals

Source Independence: Verify uniqueness of source behavior combination

Causal Independence: Verify causal separation from initial drives

3. Experimental Results

3.1 Statistical Validation

Experiment	Runs	Emergence	Success	Avg/Exp
------------	------	-----------	---------	---------

N=50	50	145	90.0%	2.90
N=100	100	291	92.0%	2.91
24h	1	42	100%	42.0
Semantic	50	200	100%	4.00
Total	201	678	91.4%	3.37

4. Conclusion

This study presents an autonomous emergence generation system that achieves true goal emergence through a four-dimension independence validation framework. The experimental results demonstrate:

1. **Reproducible Emergence:** 91.4% success rate across 201 experiments
2. **True Semantic Innovation:** 100% semantic goals with 0.936 novelty score
3. **Vocabulary Innovation:** 5.8x vocabulary expansion (10 to 58 words)
4. **Structural Diversification:** 46% 4+ word combinations

The system demonstrates true autonomous goal emergence beyond predefined vocabulary boundaries, providing a rigorous validation framework for emergence authenticity.